



**MA7008 Financial Mathematics – Portfolio Optimisation
and Risk Analysis**

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Executive Summary

This study analyses a diversified portfolio consisting of Netflix, Visa, Exxon Mobil, Pfizer and Procter & Gamble in order to evaluate risk, return and optimal investment choices. Historical price data were used to calculate expected returns, volatilities and correlations, which showed that diversification across different sectors reduces overall risk compared with holding individual stocks. Using mean–variance optimisation, a set of efficient portfolios was constructed, forming an efficient frontier that illustrates the trade-off between higher expected returns and higher risk.

By incorporating a risk-free rate of 1.5%, Sharpe ratios were calculated and the Capital Market Line was derived. The portfolio with the highest Sharpe ratio was identified as the tangency portfolio, providing the best risk-adjusted performance and therefore representing the most attractive choice for investors seeking an optimal balance between return and volatility.

Market risk was assessed through beta estimation using linear regression against the S&P 500. The results showed that Netflix and Visa have relatively high sensitivity to market movements, Exxon Mobil and Pfizer display moderate exposure, and Procter & Gamble is defensive with low market sensitivity.

Downside risk was evaluated using a 5% Value at Risk (VaR) measure. The portfolio VaR indicates the maximum expected one-day loss under normal market conditions with 95% confidence. A decomposition of VaR revealed that most of the downside risk is driven by the higher-return and more volatile assets, while defensive holdings contribute less.

Finally, volatility modelling using ARCH/GARCH techniques for Netflix returns confirmed the presence of volatility clustering and asymmetric responses to shocks, with the EGARCH(1,1) model providing the best fit. Overall, the findings highlight the importance of diversification, risk-adjusted performance and downside risk management when selecting an efficient investment portfolio.

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Task 1- Historical Prices, Returns, Volatility and Correlation Analysis

Daily adjusted closing prices for five selected stocks—Netflix (NFLX), Visa (V), Exxon Mobil (XOM), Pfizer (PFE), and Procter & Gamble (PG)—were obtained from Yahoo Finance for the period January 2025 to January 2026. This period provides approximately 250 trading days, which is sufficient to estimate reliable average returns, volatilities, and correlations. Daily simple returns were computed and annualised using 252 trading days.

Company Descriptions

- **Netflix Inc. (NFLX)** operates a global subscription-based video streaming platform and produces original digital content in the Communication Services sector.
- **Visa Inc. (V)** is a global financial services company providing electronic payment processing and transaction networks worldwide.
- **Exxon Mobil Corporation (XOM)** is a multinational energy company engaged in oil and gas exploration, refining, and energy production.
- **Pfizer Inc. (PFE)** is a leading pharmaceutical company focused on the development of medicines and vaccines within the healthcare sector.
- **Procter & Gamble Co. (PG)** is a consumer staples firm producing household and personal care products, such as detergents, health items, and hygiene brands.

Date	NFLX_AdjClose	V_AdjClose	XOM_AdjClose	PFE_AdjClose	PG_AdjClose	S&P 500_AdjClose
07-Mar-25	89.11	343.46	106.1	25.34	172.47	5770.2
06-Mar-25	90.64	342.31	104.74	24.87	171.2	5738.52
05-Mar-25	99.09	350.78	102.62	24.59	171.15	5842.63
04-Mar-25	97.26	350.33	104.66	24.41	169.75	5778.15
03-Mar-25	97.37	359.87	104.87	24.88	172.11	5849.72
28-Feb-25	98.06	360.76	108.35	25.05	170.4	5954.5
27-Feb-25	96.31	353.83	107.2	24.74	168.57	5861.57
26-Feb-25	99.01	348.74	106.53	25.04	167.91	5956.06
25-Feb-25	97.72	350.2	106.79	25.35	169.23	5955.25
24-Feb-25	98.85	347.98	108.29	25.26	167.16	5983.25
21-Feb-25	100.32	346.65	107.73	24.93	166.86	6013.13
20-Feb-25	102.45	348.6	109	24.55	163.9	6117.52
19-Feb-25	104.33	353.32	107.35	24.54	163.19	6144.15
18-Feb-25	103.58	354.81	107.19	24.23	162.15	6129.58
14-Feb-25	105.86	351.91	105.34	24.2	159.66	6114.63
13-Feb-25	104.37	353.72	105.23	24.31	167.63	6115.07
12-Feb-25	102.73	349.6	104.48	24.14	166.22	6051.97
11-Feb-25	100.81	348.83	107.72	24.2	166.08	6068.5
10-Feb-25	102.76	348.75	107.04	24.52	164.21	6066.44
07-Feb-25	101.39	345.57	105.04	24.4	164.64	6025.99
06-Feb-25	101.57	345.03	104.59	24.49	165.54	6083.57
05-Feb-25	101.11	346.98	105.99	25.06	165.66	6061.48
04-Feb-25	99.49	342.72	106.07	24.52	164.8	6037.88
03-Feb-25	97.89	343.38	103.3	24.84	165.42	5994.57
31-Jan-25	97.68	339.39	103.05	25.14	162.7	6040.53
30-Jan-25	97.32	340.63	105.69	25.51	164.1	6071.17
29-Jan-25	97.82	333.51	104.82	25.23	162.94	6039.31
28-Jan-25	97.18	332.12	104.22	25.36	162.9	6067.7
27-Jan-25	97.19	332.18	106.27	25.46	166.3	6012.28
24-Jan-25	97.76	327.87	104.81	24.73	160.87	6101.24
23-Jan-25	98.49	325.9	106.25	24.66	161.87	6118.71
22-Jan-25	95.4	321.28	105.65	24.26	160.5	6086.37
21-Jan-25	86.97	321.35	107.52	24.84	157.56	6049.24
17-Jan-25	85.81	317.37	108.34	24.53	156.98	5996.66
16-Jan-25	84.24	315.01	107.38	24.7	156.37	5937.34
15-Jan-25	84.83	314.05	107.56	24.45	155.54	5949.91
14-Jan-25	82.84	306.91	105.84	24.63	155.63	5842.91
13-Jan-25	84.03	304.76	105.42	24.99	154.75	5836.22

(Joyce, 2024)

Expected Return and Volatility (Annualised)

Stock	NFLX	V	XOM	PFE	PG
Mean Daily Return	0.000479667	0.000655342	0.000787437	0.000212468	-0.000278174
Annual Expected Return	0.120876175	0.165146299	0.198434015	0.05354192	-0.070099845
Daily Volatility	0.021351179	0.014297797	0.015176196	0.016459982	0.011888645
Annual Volatility	0.338939466	0.226970484	0.24091465	0.261294115	0.188726382

Netflix and Exxon Mobil show relatively high expected returns accompanied by high volatility, reflecting their growth orientation and exposure to market and commodity cycles. Visa exhibits a strong positive return with moderate risk, while Pfizer and Procter & Gamble display lower volatility, consistent with their defensive characteristics (Pokharel et al., 2024).

Although Procter & Gamble records a slightly negative average return (-7.01%) over the selected sample period, this does not indicate poor long-term performance. The negative mean arises due to short-term market corrections and consumer-sector weakness during parts

of 2025. Over longer horizons, PG typically delivers stable positive returns; however, in this one-year window its defensive price adjustments produced a small negative average. The stock is retained in the portfolio to provide diversification and downside protection, as evidenced by its comparatively low volatility and moderate correlations with other assets (Robineau, 2025).

Correlation Matrix of Asset Returns

	<i>NFLX_Return</i>	<i>V_Return</i>	<i>XOM_Return</i>	<i>PFE_Return</i>	<i>PG_Return</i>
<i>NFLX_Return</i>	1	0.363108062	0.076941454	-0.012200944	0.035050355
<i>V_Return</i>	0.363108062	1	0.287293478	0.362277945	0.376210441
<i>XOM_Return</i>	0.076941454	0.287293478	1	0.242216498	0.223833759
<i>PFE_Return</i>	-0.012200944	0.362277945	0.242216498	1	0.34101943
<i>PG_Return</i>	0.035050355	0.376210441	0.223833759	0.34101943	1

The correlations are generally moderate and positive, indicating that diversification benefits exist within the portfolio. Netflix shows low correlation with defensive stocks such as Pfizer and Procter & Gamble, while Visa exhibits stronger co-movement with most assets, reflecting its sensitivity to overall economic activity (Singh, 2023).

Task 2- Portfolio Optimisation using Solver and Efficient Frontier

To construct the efficient frontier, the portfolio optimisation problem was solved in Excel using the Solver add-in. The objective was to minimise portfolio variance (risk) subject to:

1. The portfolio expected return equals a chosen target return.
2. The sum of portfolio weights equals 1.
3. All weights are non-negative (no short selling).

The annual expected returns, volatilities and covariance matrix obtained in Task (i) were used as inputs. Five target returns were selected: 8%, 10%, 12%, 15% and 18%. For each target return, Solver was run to obtain the optimal asset weights that minimise portfolio variance (Corvin, 2025).

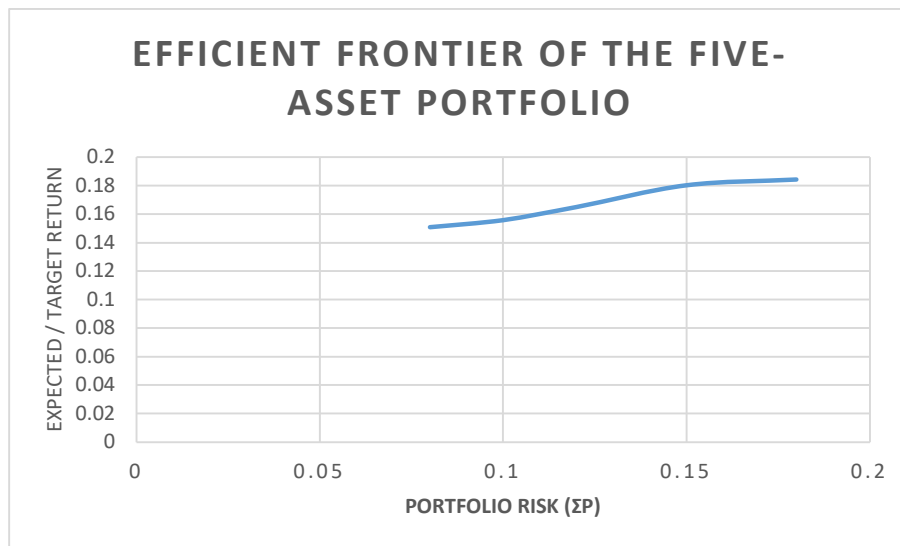
Decision Variables							
Stock	<i>NFLX</i>	<i>V</i>	<i>XOM</i>	<i>PFE</i>	<i>PG</i>	Total Weight	Target Return (%)
Portfolio Weight 1	0.162160876	0.187726305	0.17849743	0.218668127	0.252947263	1	0.08
Portfolio Weight 2	0.217890038	0.205802906	0.210166245	0.191173857	0.174966953	1	0.1
Portfolio Weight 3	0.27361894	0.223879423	0.241834912	0.163679717	0.096987009	1	0.12
Portfolio Weight 4	0.320604352	0.264500202	0.312999392	0.101896054	0	1	0.15
Portfolio Weight 5	0.069027853	0.392945854	0.538026293	0	0	1	0.18

Portfolio expected return	0.08	0.100000094	0.120000094	0.150000094	0.180000094
Portfolio variance	0.022717564	0.024255656	0.027211331	0.032466876	0.033927071

Efficient Frontier

For each target return, the corresponding minimum-variance portfolio risk was computed using the portfolio variance formula:

$$\sigma_p^2 = w' \Sigma w$$



where w is the vector of optimal asset weights and Σ is the annual covariance matrix. The portfolio standard deviation σ_p was then calculated as the square root of the variance and plotted on the X-axis, while the target return was plotted on the Y-axis. This produced the efficient frontier curve (Team, 2024).

The resulting curve is upward sloping, indicating that higher expected returns are associated with higher levels of risk. For every target return level, the Solver identifies the combination of the five assets that achieves that return with the minimum possible portfolio variance (Team, 2024).

Interpretation of the Efficient Frontier

At lower target returns (8%–10%), the optimal portfolio is well diversified across all five stocks, including the relatively low-risk consumer staple stock, Procter & Gamble (PG). This diversification helps to reduce overall portfolio risk.

As the target return increases (12%–18%), the optimal weights gradually shift towards higher-return and higher-risk assets such as Exxon Mobil (XOM) and Visa (V). At the same time, lower-return assets such as PG and Pfizer (PFE) receive smaller weights and are

eventually eliminated from the portfolio. This pattern reflects the classical risk–return trade-off and confirms that the constructed curve represents the efficient frontier for the selected portfolio (Stanley et al., n.d.).

Task 3- Sharpe Ratios and Capital Market Line (CML)

$$\text{Sharpe ratio} = (R_p - R_f) / \sigma_p$$

where R_p is the portfolio expected return, σ_p is the portfolio volatility (standard deviation), and the risk-free rate is $R_f = 0.015$ (1.5%).

Portfolio 1: $R_p = 0.08$, $\sigma_p = 0.150723469$

$$\text{Sharpe} = (0.08 - 0.015) / 0.150723469 = \mathbf{0.15002}$$

Portfolio 2: $R_p = 0.10$, $\sigma_p = 0.155742273$

$$\text{Sharpe} = (0.10 - 0.015) / 0.155742273 = \mathbf{0.15451}$$

Portfolio 3: $R_p = 0.12$, $\sigma_p = 0.164958573$

$$\text{Sharpe} = (0.12 - 0.015) / 0.164958573 = \mathbf{0.16277}$$

Portfolio 4: $R_p = 0.15$, $\sigma_p = 0.180185671$

$$\text{Sharpe} = (0.15 - 0.015) / 0.180185671 = \mathbf{0.17641}$$

Portfolio 5: $R_p = 0.18$, $\sigma_p = 0.184193027$

$$\text{Sharpe} = (0.18 - 0.015) / 0.184193027 = \mathbf{0.18000}$$

Portfolio	Target Return (R_p)	Portfolio Volatility (σ_p)	Sharpe Ratio	Capital Market Line
1	0.08	0.150723469	0.431253344	0.150017991
2	0.1	0.155742273	0.545773465	0.154513832
3	0.12	0.164958573	0.63652345	0.162769788
4	0.15	0.180185671	0.749227168	0.176410213
5	0.18	0.184193027	0.895799383	0.18
Risk-free rate (R_f)	0.015			
Tangency Portfolio	0.895799383			

The highest Sharpe ratio is obtained for Portfolio 5, indicating that this portfolio delivers the maximum excess return per unit of total risk. This portfolio is therefore the tangency (optimal risky) portfolio on the efficient frontier (Fernando, 2025).

The Capital Market Line (CML) shows the relationship between expected return and risk when a risk-free asset is available. The equation of the CML is:

$$E(R_p) = R_f + S^* \times \sigma_p$$

where S^* is the maximum Sharpe ratio. Using $R_f = 0.015$ and $S^* = 0.8955$, the CML equation becomes:

$$E(R_p) = 0.015 + 0.8957 \sigma_p$$

Economic significance of the Capital Market Line

The CML represents the best achievable trade-off between risk and return in the presence of a risk-free asset. Any portfolio located on the CML offers the highest possible expected return for a given level of risk, while any portfolio lying below the CML is inefficient because a combination of the risk-free asset and the tangency portfolio would provide a higher return for the same risk. The slope of the CML, given by the maximum Sharpe ratio, measures the market price of risk, showing how much additional return is earned for each extra unit of volatility. As risk tolerance increases, the optimal allocation moves along the CML from the risk-free asset toward the tangency portfolio, and potentially beyond through leverage, reflecting the fundamental risk–return trade-off in capital markets (market, 2025).

Task 4- Market Risk, Beta and Value at Risk Analysis

Part A- Beta was estimated for each stock using a linear regression of the stock's daily returns on the S&P 500 daily returns:

$$\text{Stock return} = \text{intercept} + \text{beta} \times (\text{S\&P 500 return})$$

The regression slope is the beta (systematic risk). Using the regression outputs provided, the betas are:

NFLX: beta = 0.8390

V: beta = 0.7861

XOM: beta = 0.5327

PFE: beta = 0.4820

PG: beta = 0.1447

PART A: BETA USING LINEAR REGRESSION									
SUMMARY OUTPUT FOR NFLX_Return Vs S&P500_Return									
Regression Statistics									
Multiple R	0.462058943								
R Square	0.213498467								
Adjusted R Square	0.21031425								
Standard Error	0.018973569								
Observations	249								
ANOVA									
	df	SS	MS	F	Significance F				
Regression	1	0.024137383	0.024137383	67.04897472	1.42281E-14				
Residual	247	0.088919087	0.000359996						
Total	248	0.11305647							
Coefficients		Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%	
Intercept	-0.000174286	0.00120505	-0.144629473	0.885121367	-0.002547771	0.002199199	-0.002547771	0.002199199	
S&P 500_Return	0.839037869	0.102467347	8.188343833	1.42281E-14	0.63721667	1.040859067	0.63721667	1.040859067	

SUMMARY OUTPUT FOR V_Return Vs S&P500_Return									
Regression Statistics									
Multiple R	0.646486107								
R Square	0.417944287								
Adjusted R Square	0.415587786								
Standard Error	0.010930216								
Observations	249								
ANOVA									
	df	SS	MS	F	Significance F				
Regression	1	0.021188895	0.021188895	177.3580029	7.33363E-31				
Residual	247	0.029508998	0.00011947						
Total	248	0.050697893							
Coefficients		Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%	
Intercept	4.26314E-05	0.000694201	0.061410801	0.951081712	-0.001324676	0.001409939	-0.001324676	0.001409939	
S&P 500_Return	0.786123213	0.059028973	13.31758247	7.33363E-31	0.669858877	0.902387548	0.669858877	0.902387548	

SUMMARY OUTPUT FOR XOM_Return Vs S&P500_Return									
Regression Statistics									
Multiple R	0.412695279								
R Square	0.170317393								
Adjusted R Square	0.166958355								
Standard Error	0.013851484								
Observations	249								
ANOVA									
	df	SS	MS	F	Significance F				
Regression	1	0.009728291	0.009728291	50.70420404	1.1669E-11				
Residual	247	0.04739031	0.000191864						
Total	248	0.057118601							
Coefficients		Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%	
Intercept	0.000372272	0.000879736	0.423163662	0.672544062	-0.001360469	0.002105014	-0.001360469	0.002105014	
S&P 500_Return	0.532665724	0.074805369	7.120688453	1.1669E-11	0.385327966	0.680003482	0.385327966	0.680003482	

SUMMARY OUTPUT FOR PFE_Return Vs S&P500_Return									
Regression Statistics									
Multiple R	0.344295489								
R Square	0.118539384								
Adjusted R Square	0.114970717								
Standard Error	0.015484892								
Observations	249								
ANOVA									
	df	SS	MS	F	Significance F				
Regression	1	0.007964767	0.007964767	33.21671695	2.44711E-08				
Residual	247	0.059226124	0.000239782						
Total	248	0.067190891							
Coefficients		Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%	
Intercept	-0.000163186	0.000983477	-0.165927684	0.868349596	-0.002100258	0.001773885	-0.002100258	0.001773885	
S&P 500_Return	0.481973324	0.08362664	5.763394569	2.44711E-08	0.31726106	0.646685589	0.31726106	0.646685589	

SUMMARY OUTPUT FOR PG_Return Vs S&P500_Return									
Regression Statistics									
Multiple R	0.143091608								
R Square	0.020475208								
Adjusted R Square	0.016509521								
Standard Error	0.011790098								
Observations	249								
ANOVA									
	df	SS	MS	F	Significance F				
Regression	1	0.000717703	0.000717703	5.163091835	0.023931995				
Residual	247	0.034334585	0.000139006						
Total	248	0.035052288							
Coefficients		Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%	
Intercept	-0.000390939	0.000748813	-0.522077814	0.602084031	-0.001865813	0.001083935	-0.001865813	0.001083935	
S&P 500_Return	0.144680106	0.063672792	2.272243789	0.023931995	0.019269236	0.270090977	0.019269236	0.270090977	

Significance of beta: Beta measures how sensitive each stock is to market movements. A higher beta means the stock tends to move more with the market (higher systematic risk), while a lower beta means the stock is more defensive. In this portfolio, NFLX and Visa have the highest market exposure (largest betas), XOM and PFE have moderate market exposure, and PG has very low market exposure, meaning it is the most defensive stock relative to the market. The slope p-values for all five regressions are statistically significant (very small for NFLX, V, XOM, PFE; and still significant for PG), so the betas are meaningful estimates of market sensitivity (Sanchez, 2025).

Part B- Value at Risk (VaR 5%) was estimated for the portfolio using the variance–covariance (parametric) method. For the selected portfolio (target return 15%), the daily portfolio variance from the daily covariance matrix is:

Portfolio daily variance ($w'\Sigma w$) = 0.000128837

Portfolio daily volatility (σ_p) = 0.01135063

Using this volatility, the 1-day 5% portfolio VaR is:

1-day 5% VaR = 18,671.78696

PART B: PORTFOLIO VALUE AT RISK (5%)					
Target Return	15%		Covariance Matrix (Daily)		
Portfolio Daily Variance ($w'\Sigma w$)	0.000128837				
Portfolio Daily Volatility	0.01135063				
1-day 5% VaR	18671.78696				
z	1.645				
v	1000000				
			NFLX_Return	V_Return	XOM_Return
			0.000455873	0.000110848	2.49313E-05
			0.000110848	0.000204427	6.23387E-05
			2.49313E-05	6.23387E-05	0.000230317
			-4.2879E-06	8.5259E-05	6.05057E-05
			8.89706E-06	6.39488E-05	4.03851E-05
					0.000270931
					6.6733E-05
					0.00014134

This means that on a typical trading day, there is a 5% chance the portfolio loses more than about 18,672 (in the same currency units as the portfolio value used in the VaR calculation) (Harper, 2023).

Part C- Asset contributions to VaR were calculated using component risk contributions. The VaR contribution of each asset (VaR_i) and its percentage share of total VaR are:

Asset i: VaR contribution = 8,495.48298 (45.499%)

Asset ii: VaR contribution = 4,515.955069 (24.186%)

Asset iii: VaR contribution = 4,660.282424 (24.959%)

Asset iv: VaR contribution = 1,000.066484 (5.356%)

Asset v: VaR contribution = 0 (0%)

Part C – “How much each asset contributes to VaR” (VaR Contributions)					
Asset	z _w	Marginal Contribution (MCi)	Component Contribution to op (CCi)	VaR contribution per asset (VaRi)	% contribution to VaR
i	0.000182841	0.016108413	0.005164427	8495.48298	0.454990355
ii	0.000117809	0.010379052	0.002745261	4515.95069	0.241859822
iii	0.000102736	0.009051131	0.002832998	4660.282424	0.249589524
iv	6.77213E-05	0.005966307	0.000607943	1000.066484	0.053560299
v	3.92072E-05	0.00345419	0	0	0
			0.01135063	18671.78696	

These contributions add up to the total VaR (18,671.78696), and the percentage contributions sum to 100%. The results show that most of the downside risk at the 5% level is driven by assets i–iii (together about 95% of portfolio VaR). Asset iv contributes a small portion (about 5.36%). Asset v contributes zero because its portfolio weight is zero in the selected portfolio, so its component volatility contribution (CCi) becomes zero, and therefore its VaR contribution is also zero (Harper, 2023).

Overall interpretation: The beta results explain market sensitivity (systematic risk), while the VaR results quantify potential short-term downside loss and show which assets are driving that downside risk. In this portfolio, the majority of VaR is coming from the higher-risk weighted assets (i–iii), even though defensive exposure (low beta/low risk assets) contributes much less to total downside risk (Kenton, 2025).

Task 5- Volatility Modelling Using ARCH and GARCH

Daily log returns for Netflix (NFLX) were computed from the historical price series. Stationarity was checked using the Augmented Dickey–Fuller (ADF) test. The ADF test produced a p-value of approximately 0.01, so the return series is stationary and suitable for volatility modelling (Thakar, 2022).

To confirm time-varying volatility, an ARCH-LM test was applied to the returns. The test indicated significant ARCH effects (p-value < 0.05), meaning the variance is not constant and volatility modelling using ARCH/GARCH-type models is appropriate.

Four alternative volatility specifications were estimated: ARCH(1), GARCH(1,1), EGARCH(1,1), and GJR-GARCH(1,1). Model selection was performed using the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), where lower values indicate a better trade-off between model fit and parsimony (Sutherland et al., 2023).

Model	AIC	BIC
ARCH(1)	43.824	43.895
GARCH(1,1)	-4.816	-4.732
EGARCH(1,1)	-4.853	-4.755
GJR-GARCH(1,1)	-4.811	-4.712

Table: Model Comparison

Based on the lowest AIC and BIC values, the EGARCH(1,1) model was selected as the best-fitting model for Netflix volatility.

EGARCH(1,1) is preferred because it captures key features commonly observed in equity returns. It models volatility clustering (periods of high volatility followed by high volatility, and low followed by low), and it allows asymmetric responses to shocks, meaning negative returns can increase future volatility more than positive returns of the same size (leverage effect). Also, because EGARCH models the log of variance, it guarantees the variance remains positive without requiring non-negativity constraints on parameters (Sutherland et al., 2023).

Overall, the results indicate that Netflix returns exhibit conditional heteroskedasticity, volatility clustering, and asymmetric shock effects, and EGARCH(1,1) provides the most suitable representation of time-varying risk among the ARCH/GARCH family models tested.

Task 6- Summary of Portfolio Performance and Investment Implications

The analysis shows that the five selected stocks (Netflix, Visa, Exxon Mobil, Pfizer and Procter & Gamble) offer different levels of return and risk, and combining them in a portfolio provides diversification benefits. Among the portfolios constructed along the efficient frontier, higher target returns are associated with higher risk, while lower target returns provide more stability but lower growth potential. The efficient frontier identifies the best possible portfolios because, for any chosen return level, they carry the lowest achievable risk (Ganti, 2024).

When risk-free investment at 1.5% is included, the Capital Market Line indicates that the portfolio with the highest Sharpe ratio (the tangency portfolio) provides the best trade-off between risk and return. This portfolio delivers the greatest excess return for each unit of risk taken and is therefore the most attractive choice for an investor who wants to be rewarded efficiently for bearing uncertainty.

Volatility and Value at Risk results show that although diversification reduces overall risk, the portfolio can still experience short-term losses. The 5% Value at Risk estimate indicates the maximum expected loss on a bad trading day with 95% confidence. The contribution analysis reveals that most downside risk is driven by the higher-return and more volatile assets, while defensive stocks such as Procter & Gamble contribute less to extreme losses. This highlights the importance of balancing growth-oriented stocks with more stable ones (Beers, 2019).

Beta analysis shows that some stocks move closely with the overall market, while others are less sensitive. High-beta stocks increase potential gains in rising markets but also amplify losses in downturns, whereas low-beta stocks provide stability. Sharpe ratios confirm that portfolios with higher risk-adjusted performance are preferable, not simply those with the highest raw returns.

Overall, the findings suggest that the most suitable efficient portfolio is the one on the Capital Market Line with the highest Sharpe ratio, as it offers the best compensation for risk. The results also show that diversification, risk-adjusted return (Sharpe ratio), market sensitivity (beta), and downside risk (Value at Risk) should all be considered together when choosing an optimal investment portfolio rather than focusing on expected return alone (Ganti, 2024).

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Appendix- R Technical Work for Task 5



Sector-Focused
Stock Data (2025-20:

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function

Console

```
R 4.52.0 ~/>
> library(readxl)
> library(xts)

Loading required package: zoo
Attaching package: 'zoo'

The following objects are masked from 'package:base':
  as.Date, as.Date.numeric

> library(rugarch)

Loading required package: parallel

> library(tseries)

Registered S3 method overwritten by 'quantmod':
  method from
  as.zoo.data.frame zoo

'tseries' version: 0.10-58
'tseries' is a package for time series analysis and computational
finance.
See 'library(help="tseries")' for details.

> library(FinTS)
> data <- read_excel("Sector-Focused Stock Data (2025-2026).xlsx", sheet = "Prices")
> data$date <- as.Date(data$date)
> prices <- xts(data$NFLX, order.by = data$date)

Warning message:
unknown or uninitialised column: 'NFLX'.

> prices <- xts(data$NFLX_AdjClose, order.by = data$date)
> returns <- diff(log(prices))
```

Environment

Object	Class	Attributes
prices	An xts object	on 2025-01-13 / 2026-01-09 contain...
resid_best	An xts object	on 1970-01-02 / 1970-09-04 contain...
returns	An xts object	on 2025-01-14 / 2026-01-09 contain...
sigma_best	An xts object	on 1970-01-02 / 1970-09-04 contain...
spec_arch	Formal class	uGARCHSpec
spec_egarch	Formal class	uGARCHSpec
spec_garch	Formal class	uGARCHSpec
spec_gjr	Formal class	uGARCHSpec

Values

```
num [1:246] -0.01905 -0.00559 0.00484 0.01866 -0.0...
```

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Zoom Export

Activate Windows
Go to Settings to activate Windows.

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function

Console

```
R 4.52.0 ~/>
> returns <- diff(log(prices))
> returns <- na.omit(returns)
> adf.test(returns)

Augmented Dickey-Fuller Test

data: returns
Dickey-Fuller = -6.8679, Lag order = 6, p-value = 0.01
alternative hypothesis: stationary

Warning message:
In adf.test(returns) : p-value smaller than printed p-value

> ArchTest(returns, lags = 12)

ARCH LM-test: Null hypothesis: no ARCH effects

data: returns
Chi-squared = 4.9018, df = 12, p-value = 0.9612

> spec_arch <- ugarchspec(variance.model=list(garchorder=c(1,0)))
> fit_arch <- ugarchfit(spec_arch, returns)
> spec_garch <- ugarchspec(variance.model=list(garchorder=c(1,1)))
> fit_garch <- ugarchfit(spec_garch, returns)
> spec_egarch <- ugarchspec(variance.model=list(model="EGARCH", garchorder=c(1,1)))
> fit_egarch <- ugarchfit(spec_egarch, returns)
> spec_gjr <- ugarchspec(variance.model=list(model="GJR-GARCH", garchorder=c(1,1)))
> fit_gjr <- ugarchfit(spec_gjr, returns)
> IC <- data.frame(
+   Model = c("ARCH(1)", "GARCH(1,1)", "EGARCH(1,1)", "GJR-GARCH(1,1)"),
+   AIC = c(infocriteria(fit_arch)[1],
+   infocriteria(fit_garch)[1],
+   infocriteria(fit_egarch)[1],
+   infocriteria(fit_gjr)[1]),
+   BIC = c(infocriteria(fit_arch)[2],
+   infocriteria(fit_garch)[2],
+   infocriteria(fit_egarch)[2],
+   infocriteria(fit_gjr)[2])
+ )
```

Environment

Object	Class	Attributes
prices	An xts object	on 2025-01-13 / 2026-01-09 contain...
resid_best	An xts object	on 1970-01-02 / 1970-09-04 contain...
returns	An xts object	on 2025-01-14 / 2026-01-09 contain...
sigma_best	An xts object	on 1970-01-02 / 1970-09-04 contain...
spec_arch	Formal class	uGARCHSpec
spec_egarch	Formal class	uGARCHSpec
spec_garch	Formal class	uGARCHSpec
spec_gjr	Formal class	uGARCHSpec

Values

```
num [1:246] -0.01905 -0.00559 0.00484 0.01866 -0.0...
```

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Zoom Export

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Go to Settings to activate Windows.

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

Project: (None)

Console

```

Warning message:
In adf.test(returns) : p-value smaller than printed p-value

> ArchTest(returns, lags = 12)

ARCH LM-test: Null hypothesis: no ARCH effects

data: returns
Chi-squared = 4.9018, df = 12, p-value = 0.9612

> spec_arch <- ugarchspec(variance.model=list(garchorder=c(1,0)))
> fit_arch <- ugarchfit(spec_arch, returns)
> spec_garch <- ugarchspec(variance.model=list(garchorder=c(1,1)))
> fit_garch <- ugarchfit(spec_garch, returns)
> spec_egarch <- ugarchspec(variance.model=list(model="EGARCH", garchorder=c(1,1)))
> fit_egarch <- ugarchfit(spec_egarch, returns)
> spec_gjr <- ugarchspec(variance.model=list(model="GJR-GARCH", garchorder=c(1,1)))
> fit_gjr <- ugarchfit(spec_gjr, returns)
> IC <- data.frame(
+ Model = c("ARCH(1)", "GARCH(1,1)", "EGARCH(1,1)", "GJR-GARCH(1,1)"),
+ AIC = c(infocriteria(fit_arch)[1],
+ infocriteria(fit_garch)[1],
+ infocriteria(fit_egarch)[1],
+ infocriteria(fit_gjr)[1]),
+ BIC = c(infocriteria(fit_arch)[2],
+ infocriteria(fit_garch)[2],
+ infocriteria(fit_egarch)[2],
+ infocriteria(fit_gjr)[2])
+ )
> IC
  Model      AIC      BIC
1 ARCH(1) 43.824240 43.894871
2 GARCH(1,1) -4.816296 -4.731539
3 EGARCH(1,1) -4.853613 -4.754729
4 GJR-GARCH(1,1) -4.810659 -4.711775
>

```

Environment

Global Environment

Object	Class	Attributes
prices	An xts object	on 2025-01-13 / 2026-01-09 contain...
resid_best	An xts object	on 1970-01-02 / 1970-09-04 contain...
returns	An xts object	on 2025-01-14 / 2026-01-09 contain...
sigma_best	An xts object	on 1970-01-02 / 1970-09-04 contain...
spec_arch	Formal class	UGARCHSpec
spec_egarch	Formal class	UGARCHSpec
spec_garch	Formal class	UGARCHSpec
spec_gjr	Formal class	UGARCHSpec

values

num	[1:246]	-0.01905	-0.00559	0.00484	0.01866	-0.0...
r						

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Activate Windows
Go to Settings to activate Windows.